The effect of coarticulation on L2 spoken word recognition

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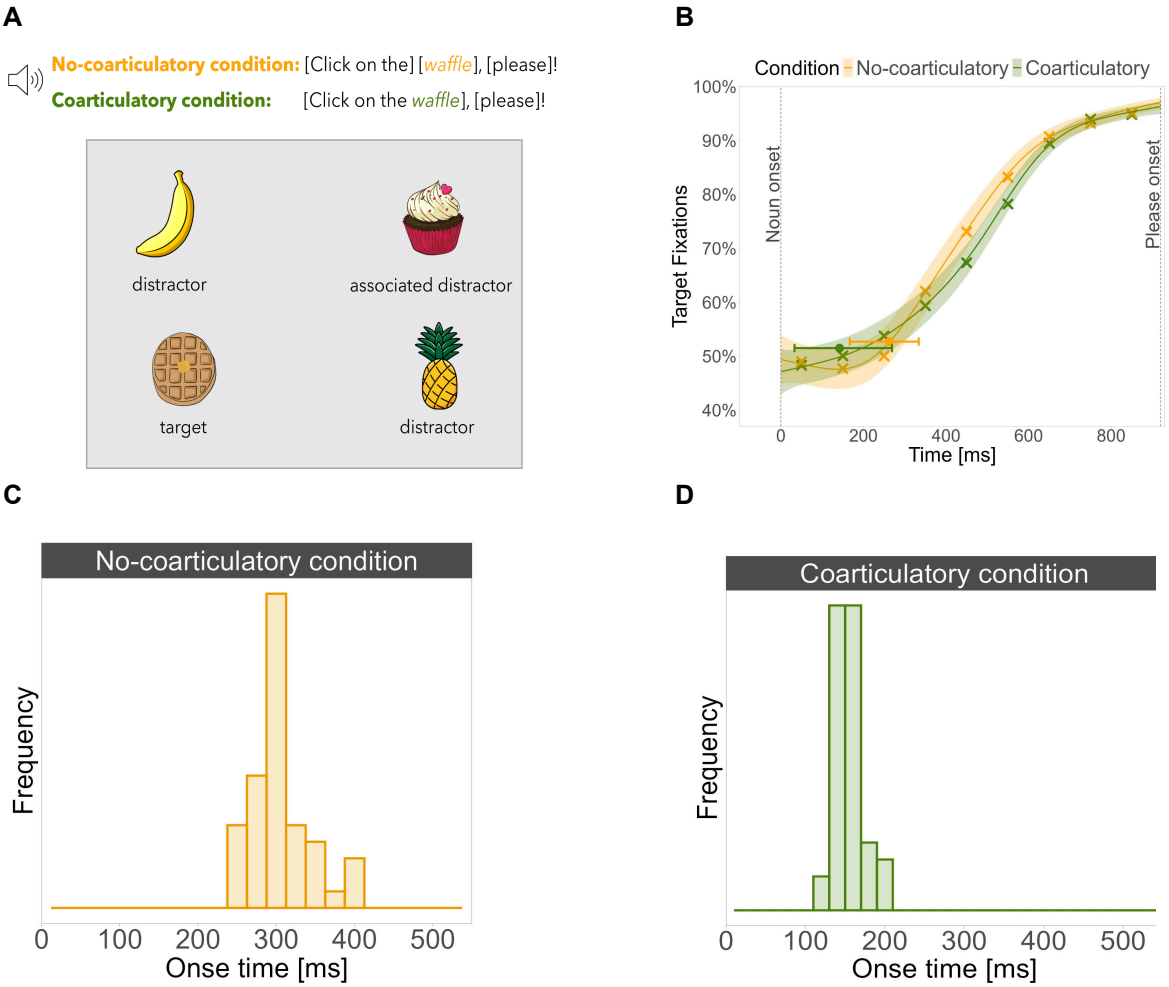
To understand speech, comprehenders rely on multiple phonetic cues in the acoustic signal as it unfolds. One primary source of phonetic information is coarticulation, a fine phonetic detail in which each sound is influenced by the surrounding sounds [1]. Visual world eye-tracking studies have shown that children and adults make use of coarticulatory features to predict upcoming linguistic elements [2,3]. However, little is known about whether second language (L2) learners—who have been shown to have fuzzy lexical and phonological representations in their L2 [4,5]—are similarly sensitive to these fine-grained coarticulatory cues. In our study, we tested whether L2 learners of English make use of coarticulatory cues in spoken word recognition, through a conceptual replication of [3]. Crucially, we investigate whether the presence of coarticulation preceding the noun results in faster spoken word recognition. To address this, we examined the earliest time point at which participants start looking at the referent in a visual world eye-tracking task. We predicted that if L2 speakers can use coarticulatory cues predictively, this time point should occur earlier in sentences containing coarticulatory information than in sentences without such information.

Methods. Forty-five German learners of English (mean age of English acquisition = 8.13 years, SD = 2.09; LexTALE mean score = 80.3%, SD = 12.4) performed a visual world eye-tracking experiment. In each trial, participants heard instructions like “Click on the waffle, please!” while looking at displays with four objects: the target object, an associated distractor, and two other distractors (Fig. 1A). In the no-coarticulatory condition, the determiner and noun were recorded separately and cross-spliced to eliminate coarticulatory cues in the determiner. In the coarticulatory condition, the determiner and noun constituted a continuous phonological phrase. Following [3], a Gaussian classifier was used to demonstrate that coarticulatory information was present in the determiner’s vowel. The eye-tracking data were analyzed with a GAMM-based method [6] that allowed estimating the earliest time point at which looks to the target were significantly greater than to the associated distractor, i.e., the onset. Participants also completed a demographic questionnaire and the English vocabulary test LexTALE, which has been shown to correlate with L2 proficiency [7].

Results and discussion. At the group level, the onset for the no-coarticulatory condition was estimated at 263 95% CI [164, 336] ms, and for the coarticulatory condition at 144 ms 95% CI [38, 275] ms (Fig. 1B). The mean difference between the two conditions was 115 95% CI [-46, 259] ms. Although the difference was not statistically significant, as the confidence interval included zero, there was a clear tendency for L2 speakers to exhibit shorter onsets in the coarticulatory condition (data collection is still ongoing; target sample according to power analysis = 90). This was further supported by the by-participant onsets: We observed higher inter-individual variability in the no-coarticulatory condition than in the coarticulatory condition (Fig. 1C–D). Moreover, we plan to perform individual differences analyses, i.e., age of acquisition and proficiency, once the full dataset is collected. Overall, our findings indicate that, as reported for native speakers [3], L2 learners recruit language-specific patterns of phonetic organization, such as coarticulation, to support spoken word recognition in their L2. This

contributes to a broader understanding of the extent to which L2 learners encode and use fine phonetic detail during language processing, indicating they learn position-specific allophones in ways that parallel L1 representations [3].

Figure 1. (A) Sample visual display in the experiment. Following [3], one of the three distractors was named “associated distractor” and used to compare with the target object. The target nouns started with an approximant (/r/, /l/, /w/) or with a voiceless fricative (/f/, /s/, /ʃ/). (B) Fitted GAMM curve overlaid onto empirical means of fixation proportions across participants. Points with error bars indicate the median speech-to-eye lag with the 95% highest density interval obtained from the GAMM-based onset detection method. (C) Onset at the participant level in the no-coarticulatory condition and (D) in the coarticulatory condition.



References: [1] Daniloff & Hammarberg (1973). *J. Phon.* [2] Cross & Joanisse (2018). *Lang. Cogn. Neurosci*. [3] Salverda et al. (2014). *JML*. [4] Bordag et al. (2021). *BLC*. [5] Bordag et al. (2022). *BLC.* [6] Veríssimo & Lago (2025). Preprint: https://doi.org/10.31234/osf.io/yk4xb_v4. [7] Lemhöfer & Broersma (2012). *Behav. Res. Methods.* [8] Flege et al. (2021). *Second language speech learning: Theoretical and empirical progress*.